Biostatistics Study Design

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WHY BIOSTATISTICS?

- Biostatisticians work at the interface of **statistics** with **biomedicine** to solve **important design and analysis questions** that are critical to the success of scientific research.
- The advent of modern biomedical technologies like **microarrays, next generation sequencers, magnetic resonance imaging, and mass spectrometry** is generating **enormous amounts** of data that has created many new challenges and opportunities for biostatisticians.
- This is an exciting field where your quantitative skills are needed to help solve **real-life scientific problems**. Due to the rapid pace that this field is growing, biostatisticians have many exciting career opportunities and are in high demand in the job market.

Statistics is the Grammar of Science.
(Karl Pearson)

Objective
- What is the rational of the study?
- What are the expected outcomes?

Study Design
- Type of study
- Sample

Sample Size
- Power analysis
- 3Rs

Experimental setup
- Groups
- Replica

Data Analysis
- Results
- Conclusion
- Outlook

Report Writing
- Paper
- Thesis

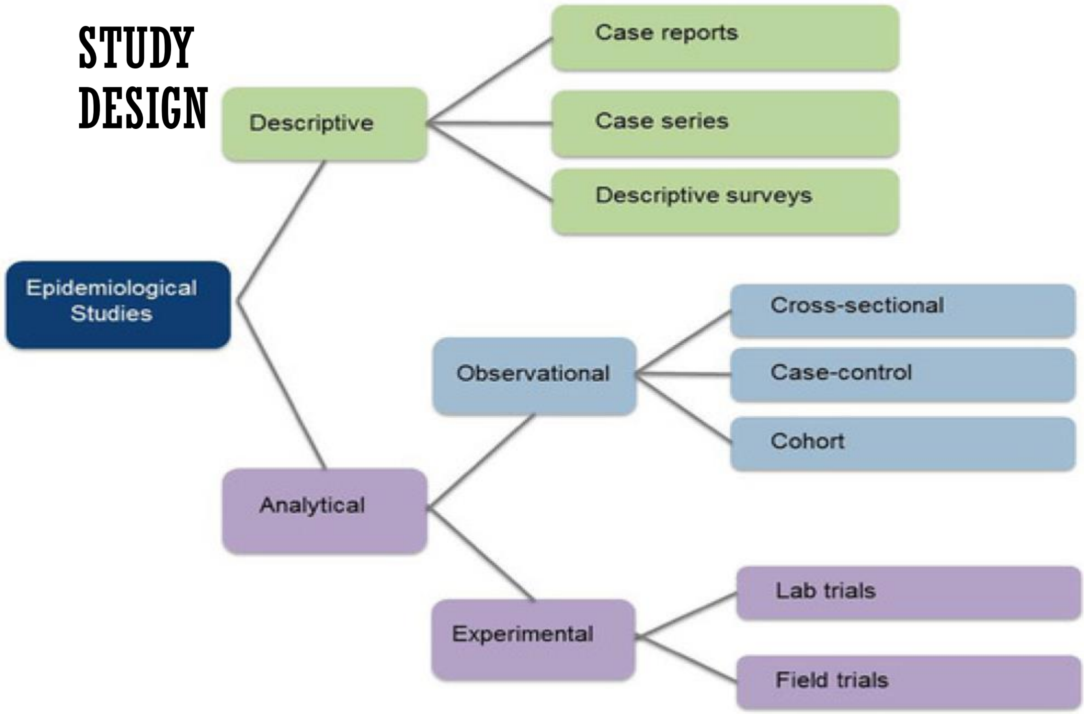
STUDY DESIGN

Study design

Experimental studies
(intervention of the researcher, observation of what happens)

Observational studies
(subjects are observed, no action from the researcher)

STUDY DESIGN

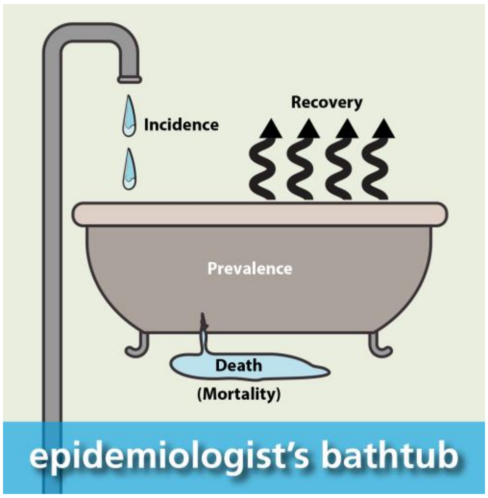


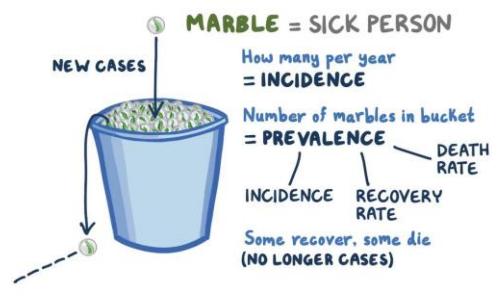
Study design	Characteristics	Strengths	Weaknesses
Case report and case series	One or a few subjects Detailed description of (a) case(s) without a control group	First form of publication Fast, inexpensive Hypothesis generating	Very limited potential to establish causal effects Selection bias*
Cross-sectional study	Exposure and outcome measured at same point in time Subjects with and without outcome are compared	Useful to describe the prevalence of disease Fast, inexpensive Hypothesis generating	Very limited potential to establish causal effects Selection bias* Survival bias*
Case-control study	Cases (those with the outcome of interest) are compared with controls (those without the outcome of interest) with respect to exposure	Efficient Suitable to study rare outcomes and multiple exposures Relatively inexpensive Hypothesis generating	Some potential to establish causal effects Can only study one outcome Choice of control group can be difficult Selection bias* Recall bias*
Cohort study	A cohort of subjects free of the outcome is followed and compared based on the exposure	Suitable to study multiple exposures, rare exposures, and multiple outcomes Hypothesis generating High generalizability	Some potential to establish causal effects Can take a long period Can be expensive Selection bias*
RCT	Randomization: allocation of subjects to experimental or control group by chance	Gold standard in establishing causal effects in studies on therapy Suitable to study more than one intervention	Very expensive Can take a long period Not suitable to study rare events Can be unethical Often low generalizability due to strict selection criteria

* Each study design may suffer from specific types of bias. These will be explained in the following papers of this series.

Cohort studies are used to study incidence, causes, and prognosis. Because they measure events in chronological order they can be used to distinguish between cause and effect.
Cross sectional studies are used to determine prevalence.

INCIDENCE & PREVALENCE





INCIDENCE & PREVALENCE

- **Prevalence**:
- Prevalence looks at existing cases, while incidence looks at new cases.
- In a population of 10,000 people, 500 persons are reported to be affected by a certain disease. So what is the prevalence of this disease in this population?
- The mathematical way to calculate this would be:

$$\frac{\textit{number of cases}}{\textit{population}} \times 100$$

- This formula will provide us with the information as a percentage. By dividing 500 by 10,000 and multiplying the result by 100 (to make it a percentage), we find out that 5% of the population is affected. So the prevalence of the disease in our population is **5%.**

INCIDENCE & PREVALENCE

- Rather than expressing prevalence as a percentage, we can also describe it as the number of people affected in a standard sized population, for example 1,000 people. So instead we would calculate:

$$\frac{500}{10{,}000} \times 1{,}000 = 50$$

In epidemiology, we actually have three different ways to calculate the prevalence:

•**Point prevalence**: The number of cases of a health event at a certain time. For example, in a survey **you would be asked if you are currently smoking.**

•**Period prevalence**: The number of cases of a health event in reference to a time period, often 12 months. For example, in a survey **you would be asked if you have smoked during the past 12 months**.

•**Lifetime prevalence**: The number of cases of the health event in reference to the total lifetime. For example, in a survey, **you would be asked if you have ever smoked.**

INCIDENCE & PREVALENCE

- **Incidence**:
- HIV is nowadays a treatable infection with a normal life expectancy. This means that with stable numbers of new cases, prevalence numbers will increase. Looking at the new cases (incidence) provides a deeper understanding of what is going on.

$$\text{Incidence} = \frac{\text{new cases}}{\text{total population}}$$

In a population of 1,000 non-diseased persons, 28 were infected with HIV over two years of observation. The incidence proportion is 28 cases per 1,000 persons, i.e. 2.8% over a two year period or 14 cases per 1,000 person-years (incidence rate), because the incidence proportion (28 per 1,000) is divided by the number of years

COHORT STUDIES

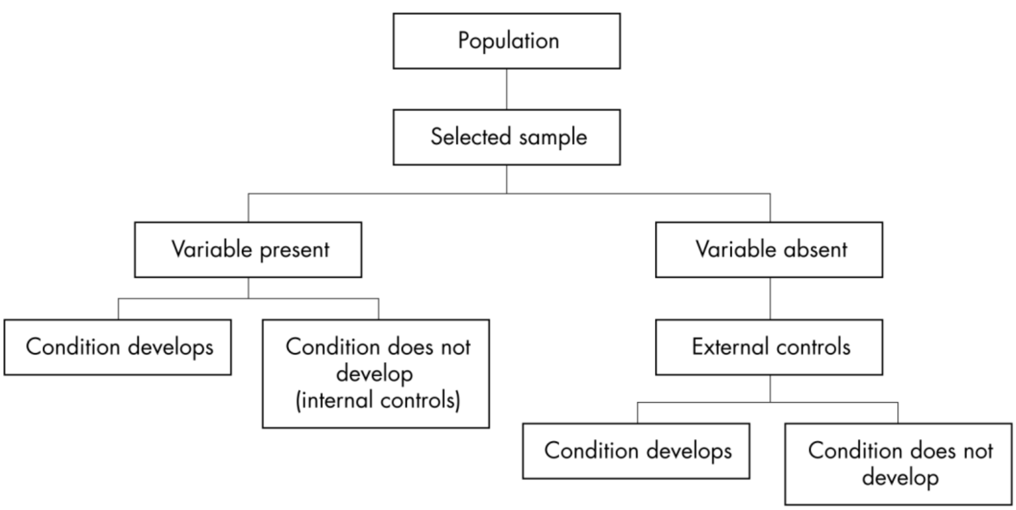
These are **the best method for determining the incidence and natural history of a condition**. The studies may be **prospective or retrospective** and sometimes two cohorts are compared.

Key points

Cohort studies

- Cohort studies describe incidence or natural history.
- They analyse predictors (risk factors) thereby enabling calculation of relative risk.
- Cohort studies measure events in temporal sequence thereby distinguishing causes from effects.
- Retrospective cohorts where available are cheaper and quicker.
- Confounding variables are the major problem in analysing cohort studies.
- Subject selection and loss to follow up is a major potential cause of bias.

Mann, C., Emerg Med J 2003;**20**:54–60



CROSS SECTIONAL STUDIES

- These are **primarily used to determine prevalence**.
- Prevalence equals **the number of cases in a population at a given point in time.**
- All the measurements on each person are made **at one point in time.**
- Prevalence is vitally important to the clinician because it influences **considerably the likelihood of any particular diagnosis and the predictive value of any investigation.**
- For example, knowing that ascending cholangitis in children is very rare enables the clinician to look for other causes of abdominal pain in this patient population.

Key points

Cross sectional studies

- Cross sectional studies are the best way to determine prevalence
- Are relatively quick
- Can study multiple outcomes
- Do not themselves differentiate between cause and effect or the sequence of events

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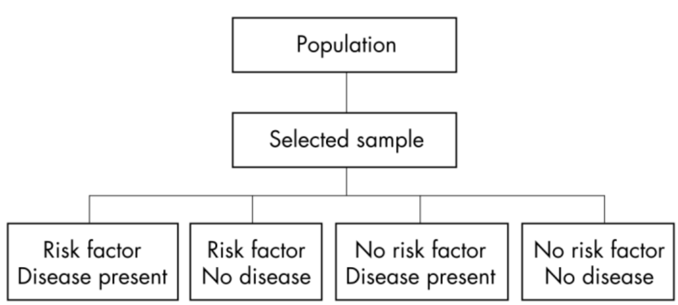
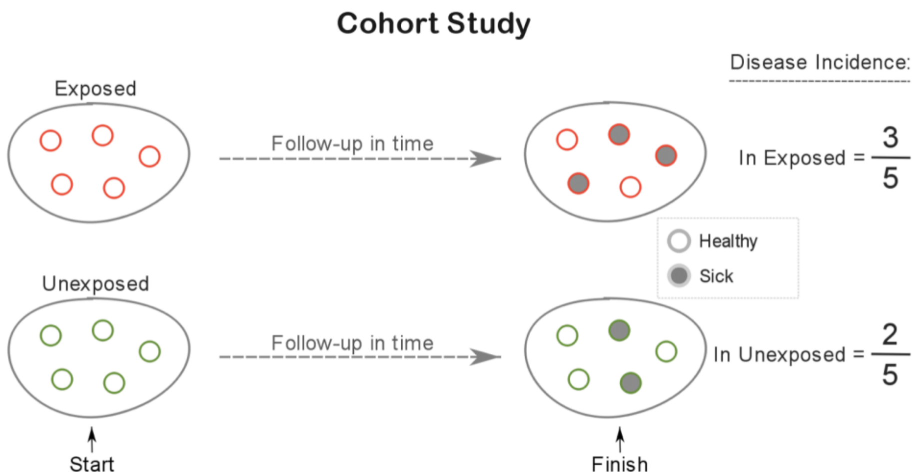
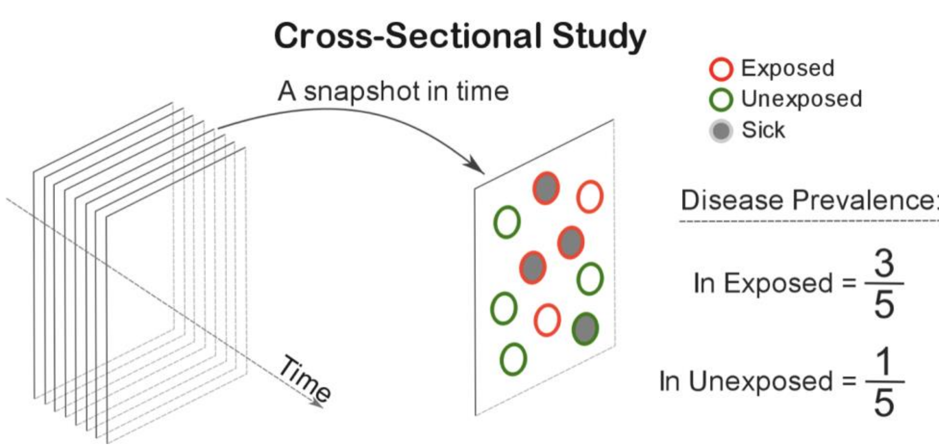


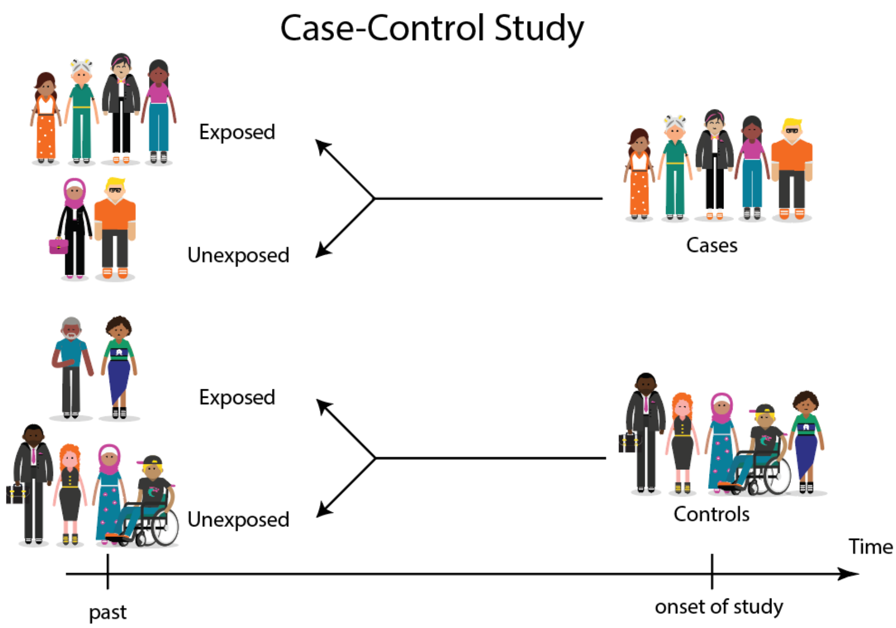
Figure 2 Study design for cross sectional studies

Mann, C., Emerg Med J 2003;**20**:54–60





In a **cross-sectional study**, the researcher collects data simultaneously on both exposure and outcome at one given point in time.



- Prospective Cohort Studies
  - Randomized/controlled study design
  - Observational (Cohort) Studies

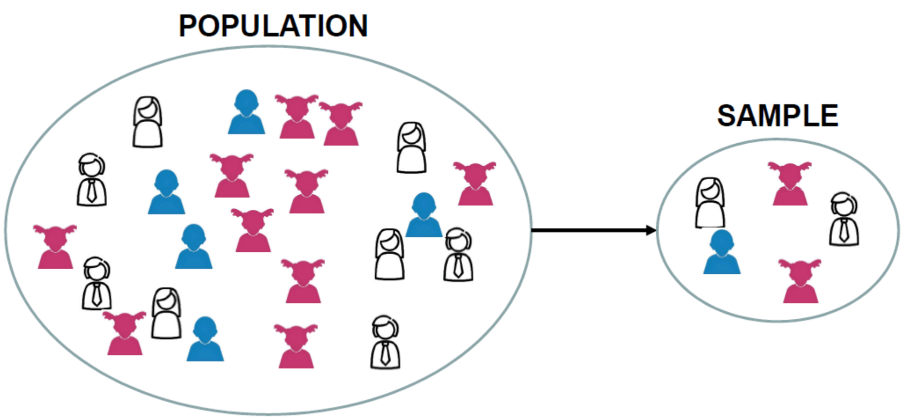
  Subjects are classified as to their exposure(s) status at study start, and followed over time to see who develops outcome(s)

- Case/Control Studies

  Subjects are chosen based on their outcome status, and the exposure(s) that occurred prior to outcome are assessed

SAMPLING

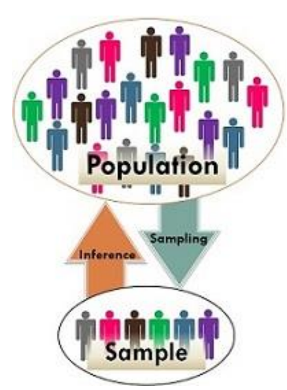
Population and Sample



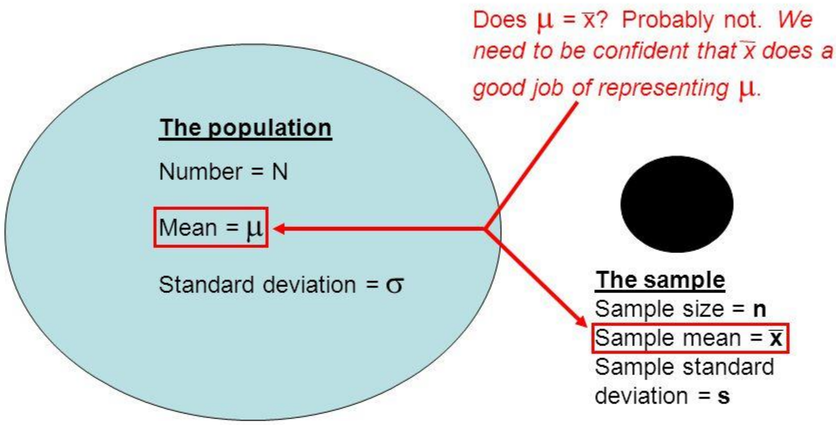
A population is the collection or set of all of the values that a variable may have.

- A sample is a part of a population.
- We use the data from the sample to make inference about the population
- The sample mean is not true mean but might be very close.
- Closeness depends on sample size

	Population	Sample
Definition	Complete enumeration of items is considered	Part of the population chosen for study
Characteristics	Parameters	Statistics
Symbols	Population size = N Population mean = μ Population S.d = σ	Sample size = n Sample mean = $\bar{x}$ Sample S.d = s

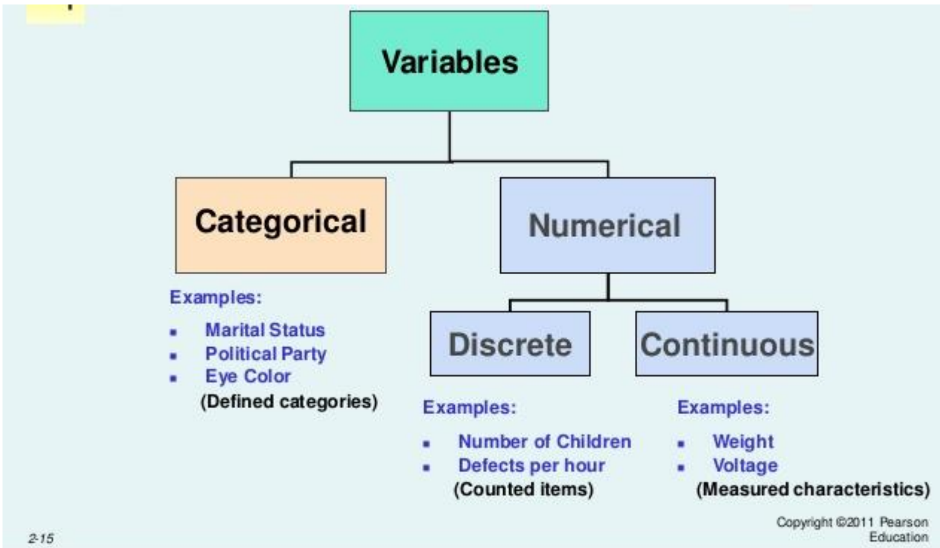


The Population vs. The Sample



Basis for Comparison	**Sample Mean**	**Population Mean**
Definition	Sample mean is the Arithmetic mean of a Small sample data drawn from a Population.	Population Mean represent the arithmetic mean of whole population.
Accuracy	It consists of lower accuracy	Its Accuracy is Higher
Formula	$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} a_I$	$\mu = \frac{1}{N} \sum_{i=1}^{N} a\,i$
	It is regarded as an efficient and unbiased estimator of population mean which means that the most expected value for the sample statistic is the population statistic, irrespective of the sampling error.	It is a mean of group characteristic, where group refers to elements of the population like items, persons, etc. and the characteristic is the item of interest. As the population is very large and not known, the population mean is unknown constant.

DATA TYPES


Variables
Categorical
Numerical
Examples:
Marital Status
Political Party
Eye Color
(Defined categories)
Discrete
Continuous
Examples:
Number of Children
Defects per hour
(Counted items)
Examples:
Weight
Voltage
(Measured characteristics)
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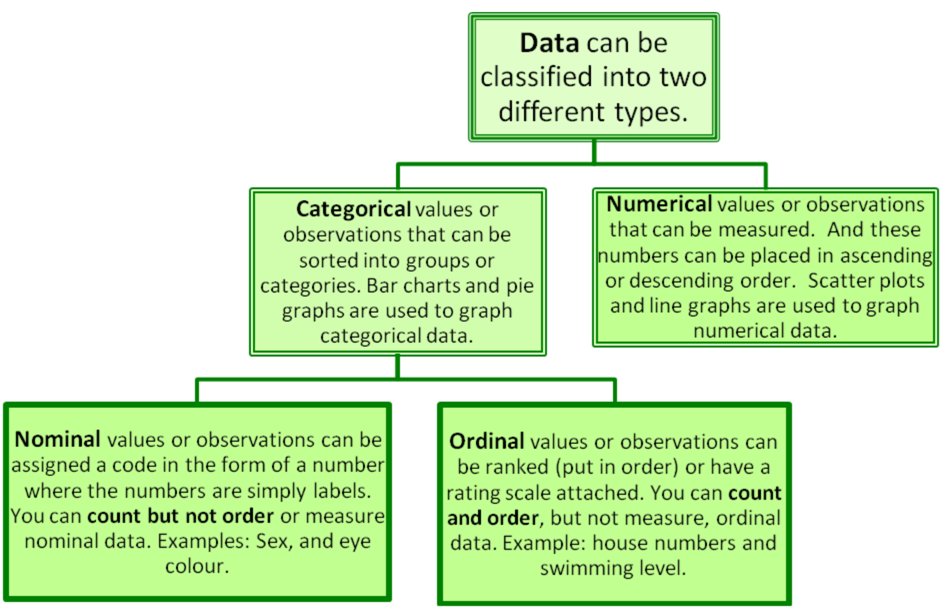

Data can be classified into two different types.
Categorical values or observations that can be sorted into groups or categories. Bar charts and pie graphs are used to graph categorical data.
Numerical values or observations that can be measured. And these numbers can be placed in ascending or descending order. Scatter plots and line graphs are used to graph numerical data.
Nominal values or observations can be assigned a code in the form of a number where the numbers are simply labels. You can **count but not order** or measure nominal data. Examples: Sex, and eye colour.
Ordinal values or observations can be ranked (put in order) or have a rating scale attached. You can **count and order**, but not measure, ordinal data. Example: house numbers and swimming level.

- Categorical data : an extension of binary data to include more than 2 possible values

- Nominal categorical data: no inherent order to categories
  - Race/ethnicity
  - Country of birth
  - Religious Affiliation

- Ordinal categorical data: order to categories
  - Income level categorized into four categories, least to greatest
  - Degree of agreement, five categories from strongly disagree to strongly agree

CONTINUOUS DATA MEASURES

Name & Meaning	Formula / Example	Used for
Arithmetic Mean [average]	$\frac{sum}{size} = \frac{a+b+c}{3}$	Most situations ("average item")
Median [middle value]	Middle of sorted list (2 middles? Average 'em)	Wildly varying samples (houses, incomes)
Mode [most popular]	Most popular value	No compromises (winner takes all)

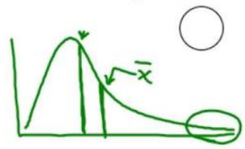
Mean

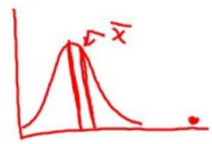
- Measure of the center of a set of data – the arithmetic average.

Pronounced "x-bar"

$$\bar{x} = \frac{Total}{n} = \frac{\Sigma x}{n}$$

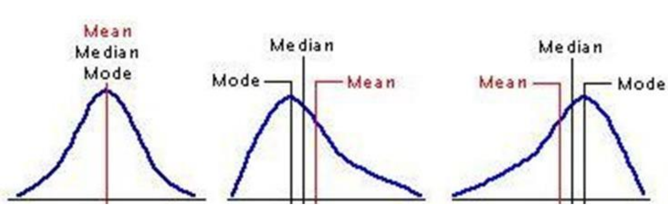
- **<u>Mean is not resistant</u>** – it is easily influenced by outliers or skewed data.

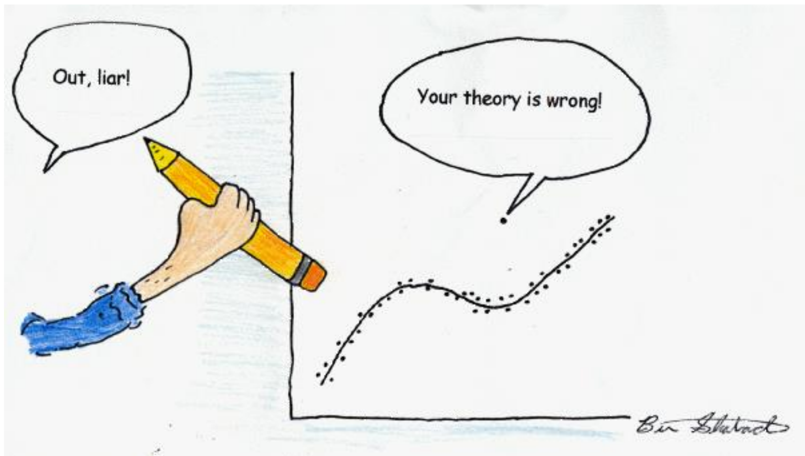




Mean vs. Median

- The relationship between the mean and median depends on the shape of the distribution
  - Symmetric – mean = median
  - Skewed-left – median < mean
  - Skewed-right – median > mean





Thank you